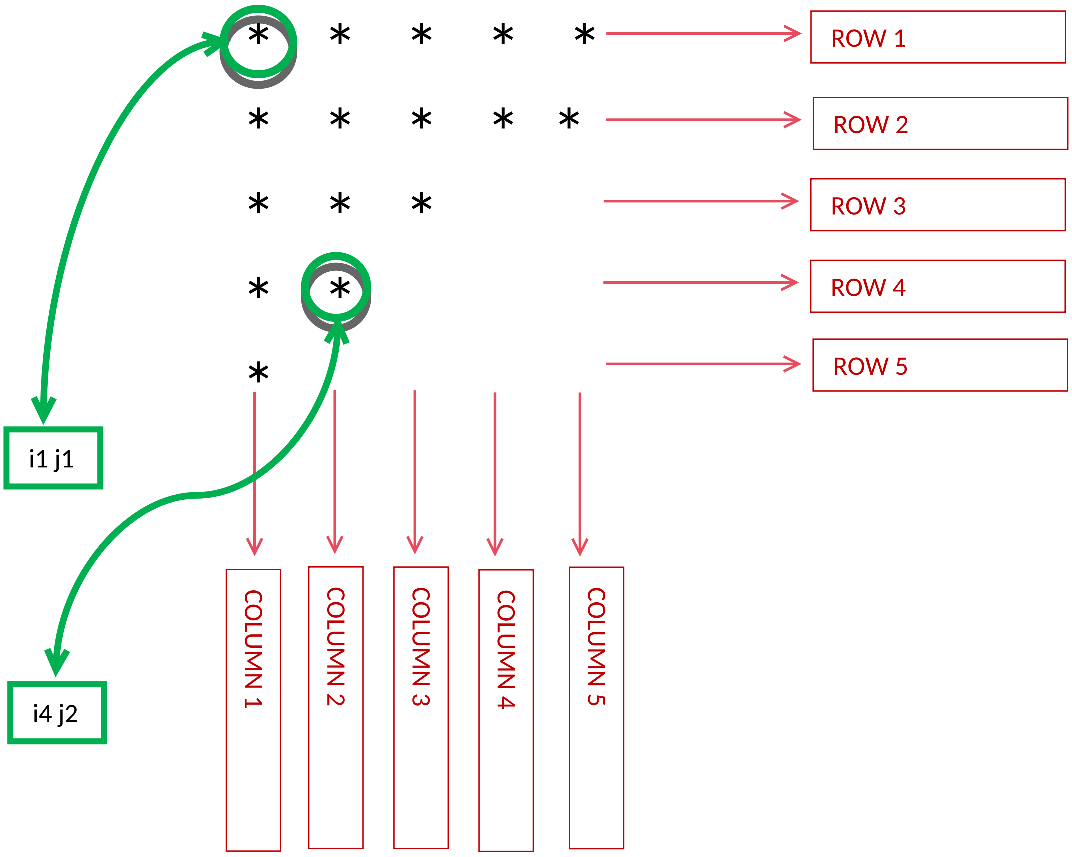
WEEK 01 MODULE
Programming Practice

<u>TOPIC : 1 ‘ Star Pattern’</u>

5. INVERTED TRIANGLE ⟵⟶ HALF DIAMOND(LOWER)



1. Problem Analysis :-

i. Pattern Structure:

- The output forms an inverted right-angled triangle.

- Unlike the Triangle Pattern where the number of stars increases in each row, this pattern decreases the number of stars by one in every subsequent row.

- The pattern starts with n stars in the first row and decreases by one star in each subsequent row until the last row contains one star.

- For n=5

```
*   *   *   *   *  →  Row 1 : (1,1) (1,2) (1,3) (1,4) (1,5)
*   *   *   *   *  →  Row 2 : (2,1) (2,2) (2,3) (2,4)
*   *   *          →  Row 3 : (3,1) (3,2) (3,3)
*   *              →  Row 4 : (4,1) (4,2)
*                  →  Row 5 : (5,1)
```

ii. Execution Flow:

<u>Step 1</u>: Observe the Grid Concept:

- The pattern can be visualized as a two-dimensional grid (matrix)

- size = n × n.

- Each cell in the grid represents a unique coordinate (i, j), where i denotes the row index and j denotes the column index.

- For n = 5:

As the row number increases by 1, the number of stars decreases by 1.
Observation:

- Row 1 → 5 star
-
- Row 2 → 4 stars
-
- Row 3 → 3 stars
-
- Row 4 → 2 stars
-
- Row 5 → 1 stars
- ......

A clear relationship can be observed: Number of stars = n - i + 1

Therefore:

- Row 1 prints n stars
- Row 2 prints n-1 stars
- Row 3 prints n-2 stars
- ......
- Row n prints 1 star

- The number of valid columns is decreasing.

- Every new row removes one column from the right side.

- Therefore the inner loop cannot remain fixed.

- its limit must shrink with every row.

Step 2: Traverse the grid row by row.

- The outer loop is responsible for generating rows.

Step 3: For every row generated by the outer loop, determine how many stars need to be printed.

- The number of stars depends on the current row number.
- The relationship is:

- Number of Stars = n - i + 1

Step 4: Traverse the required number of columns using the inner loop.

- The inner loop is responsible for generating columns and stars in the current row.
- The loop executes from:

- 0 to (n - i - 1)

Step 5: At each iteration of the inner loop, print a star (*).

Step 6: After all stars of the current row have been printed, move the cursor to the next line.

Step 7: Repeat the same process until all n rows have been completed.

2. Program :-

```
package week1_module;
import java.util.Scanner;
public class InvertedTraingle {

	public static void main(String[] args) {

		Scanner sc = new Scanner(System.in);

		System.out.print("Enter the number of rows: ");
		int n = sc.nextInt();

		for(int i = 0; i < n; i++) {

			for(int j = 0; j < n - i; j++) {
				System.out.print("* ");
			}

			System.out.println();
		}

		sc.close();
	}
}
```

3. Program Output :-

```
Enter the number of rows: 5

* * * * *
* * * *
* * *
* *
*
```

4. ScreenShot :-

i. Program :-

InvertedTraingle.java

```java
package week1_module;
import java.util.Scanner;
public class InvertedTraingle {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        System.out.print("Enter the number of rows: ");
        int n = sc.nextInt();

        for(int i = 0; i < n; i++) {

            for(int j = 0; j < n - i; j++) {
                System.out.print("* ");
            }

            System.out.println();
        }

        sc.close();
    }
}
```

ii. Program's Output :-

Problems | Javadoc | Declaration | Console

```
<terminated> InvertedTraingle [Java Application] /Library/Java/JavaVirtualMachines/temurin-25.jdk/Contents/Home/bin/java
Enter the number of rows: 5
* * * * *
* * * *
* * *
* *
*
```